Ultra dasik Algebra test

Name: _____

class: _____ Sec____ roll_____ Date of borth _____

1. Expand using an appropriate algebraic identity: $(x + 7)^2$

2. Expand: $(3a - 5b)^2$

3. Simplify: $(x + 4)^2 - (x - 4)^2$

4. Using an appropriate identity, find: 103^2

5. Expand and simplify: $(2x + 3y)^3$

6. Expand: $(a - b)^3$

7. Simplify: $(x + y)^3 - (x - y)^3$

8. Evaluate using an algebraic identity: 99^3

9. Factorise completely: $x^2 + 7x + 12$

10. Factorise:$x^2 - 9x + 20$

11. Factorise:$2x^2 + 7x + 3$

12. Factorise: $3x^2 - 11x + 6$

13. Factorise using an algebraic identity:$25a^2 - 49b^2$

14. Factorise:$4x^2 - 12x + 9$

15. Factorise completely: $6x^2 + x - 2$

16. Factorise:$x^3 + 3x^2 + 3x + 1$

17. Factorise:$a^3 - 8$

18. Factorise:$x^3 - 27y^3$

19. Factorise completely:$x^3 + 8y^3$

20. Factorise: $8a^3 - 125b^3$

21. Solve using Sridharacharya's formula: $x^2 - 5x + 6 = 0$ 22. $2x^2 - 7x + 3 = 0$ 23.$3x^2 + 5x - 2 = 0$ 24. $4x^2 - 4x - 15 = 0$ 25. $5x^2 - 13x + 6 = 0$

26. Find the degree, leading coefficient and constant term of:$7x^4 - 3x^2 + 5x - 9$

27. Add:$(3x^2 + 5x - 7) + (4x^2 - 2x + 9)$

28. Subtract $(2x^3 - 5x^2 + 3x - 4)$ from $(7x^3 + 2x^2 - 6x + 8)$

29. Find the value of the polynomial: $2x^3 - 3x^2 + 4x - 5$ at $x = 2$.

30. Prove that $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

1) $A = \pi r^2$

2)$(x+a)^n = \sum_{k=0}^{n} \binom{n}{k} x^k a^{n-k}$

3)$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L}\right)$

4)$(1+x)^n = 1 + \frac{nx}{1!} + \frac{n(n-1)x^2}{2!} + \cdots$

5)$\sin\alpha \pm \sin\beta = 2\sin\frac{1}{2}(\alpha \pm \beta)\cos\frac{1}{2}(\alpha \mp \beta)$

thank you